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Hayek et al.

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(54) **CONCRETE POST-TENSIONING ANCHORS**

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E04C 5/12 (2006.01)

(52) **U.S. Cl.**
CPC ***E04C 5/122*** (2013.01); *E04C 5/125* (2013.01)

(58) **Field of Classification Search**
CPC E04C 5/12; E04C 5/122; E04C 5/125
See application file for complete search history.

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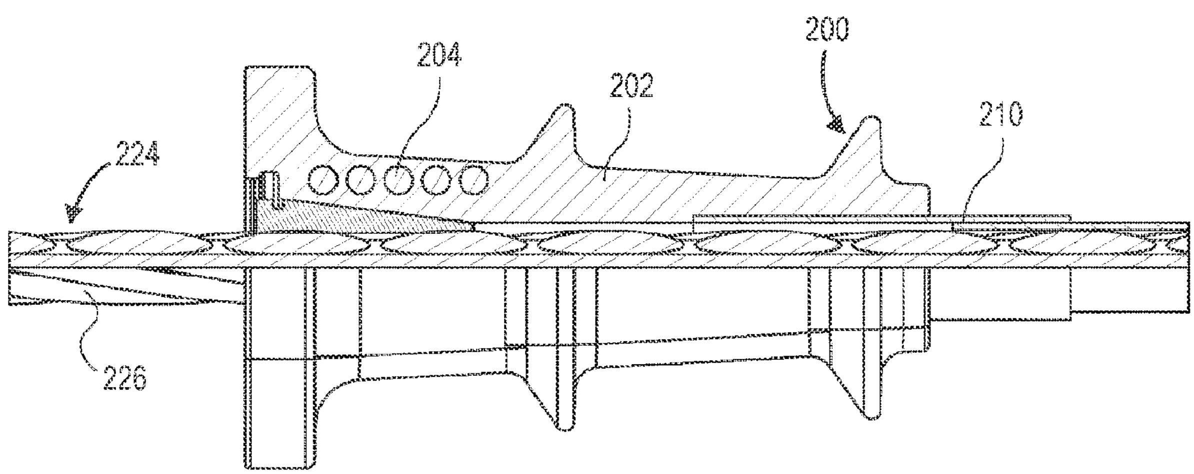
(57) **ABSTRACT**

A post-tensioning anchor is configured to post-tension at least one tension steel element. The post-tensioning anchor includes a force transfer unit configured to transmit pre-stressing force into a surrounding concrete substrate and at least one steel member configured to resist an anchoring force by the at least one tension steel element on the force transfer unit. The force transfer unit is made of concrete.

20 Claims, 13 Drawing Sheets



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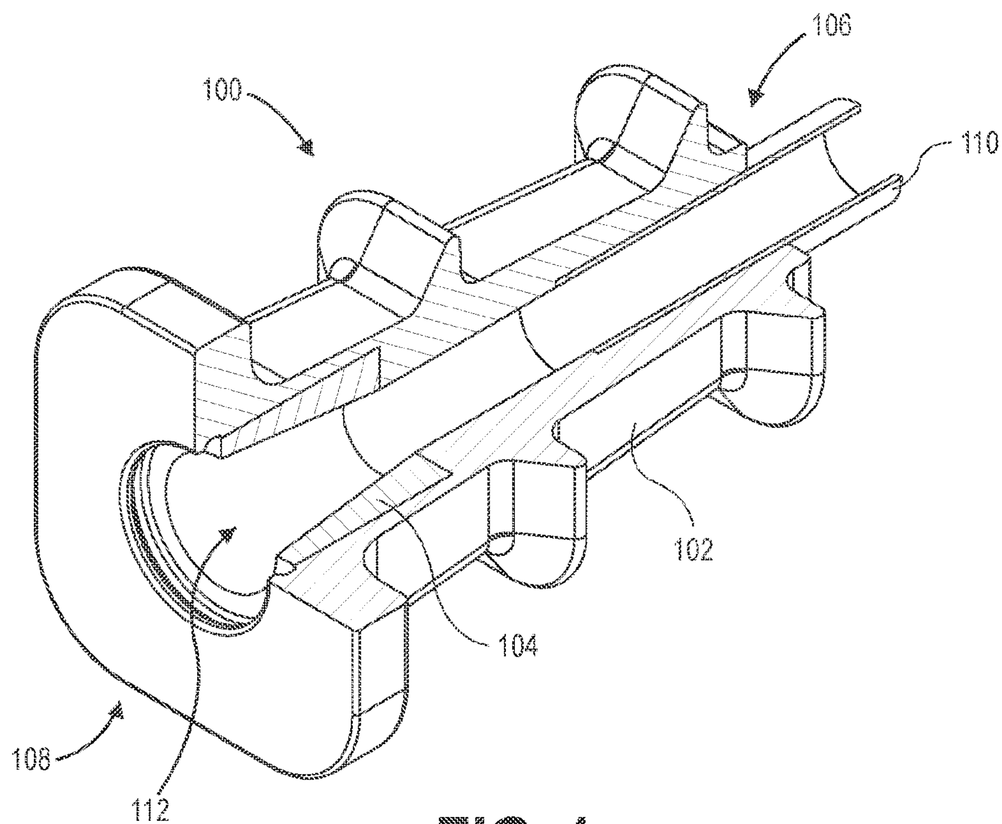

FIG. 1

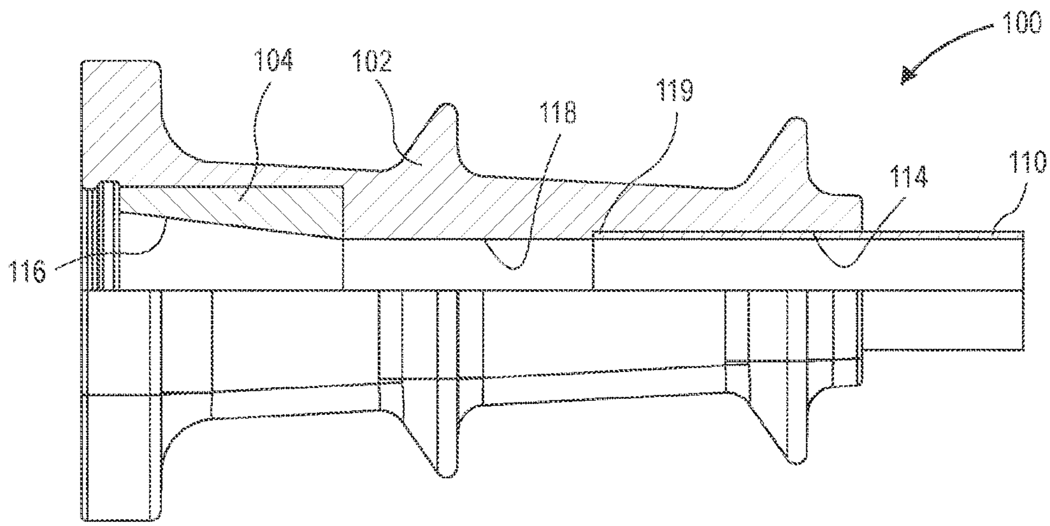

FIG. 2A

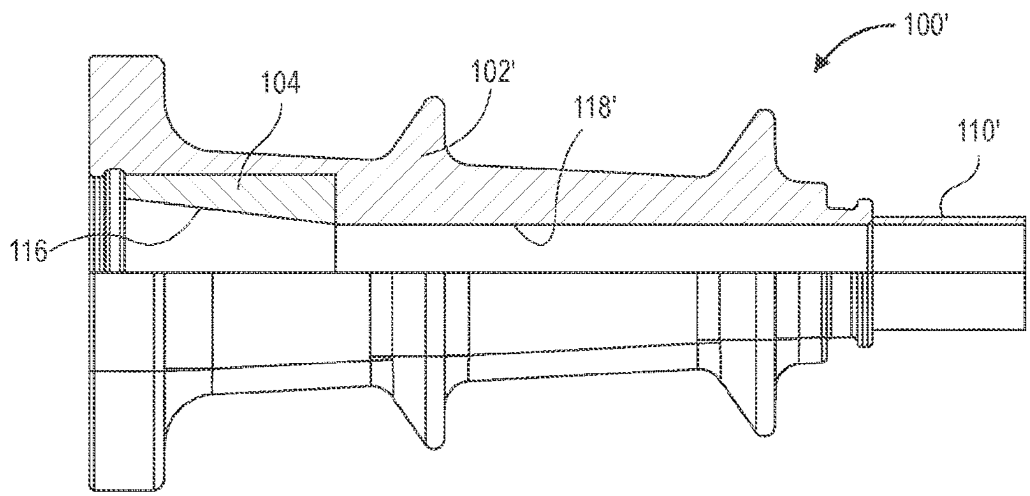

FIG. 2B

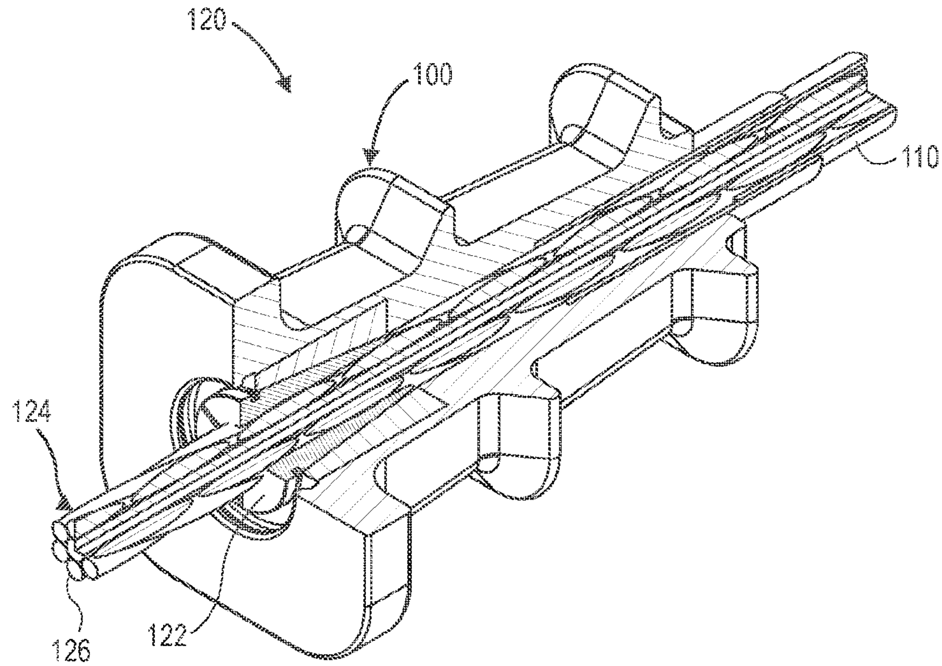

FIG. 3

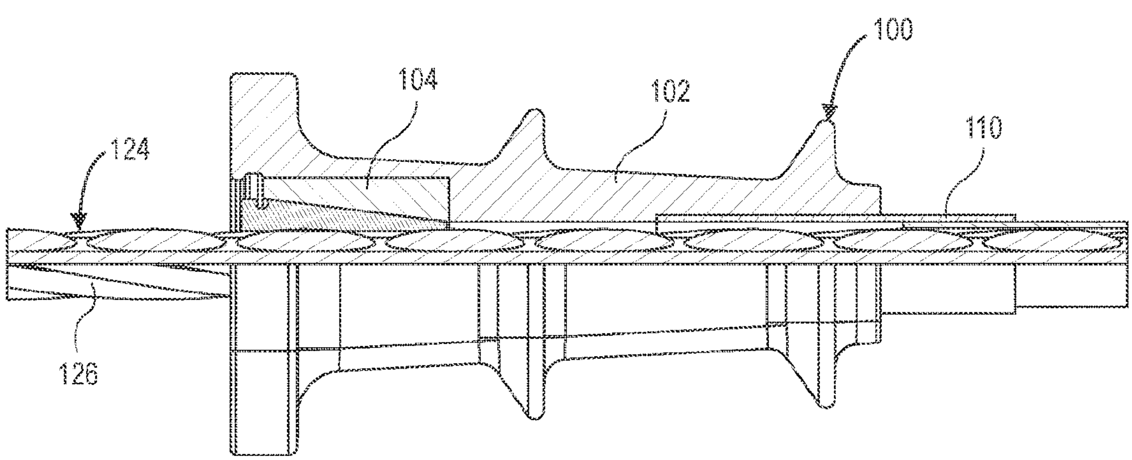

FIG. 4

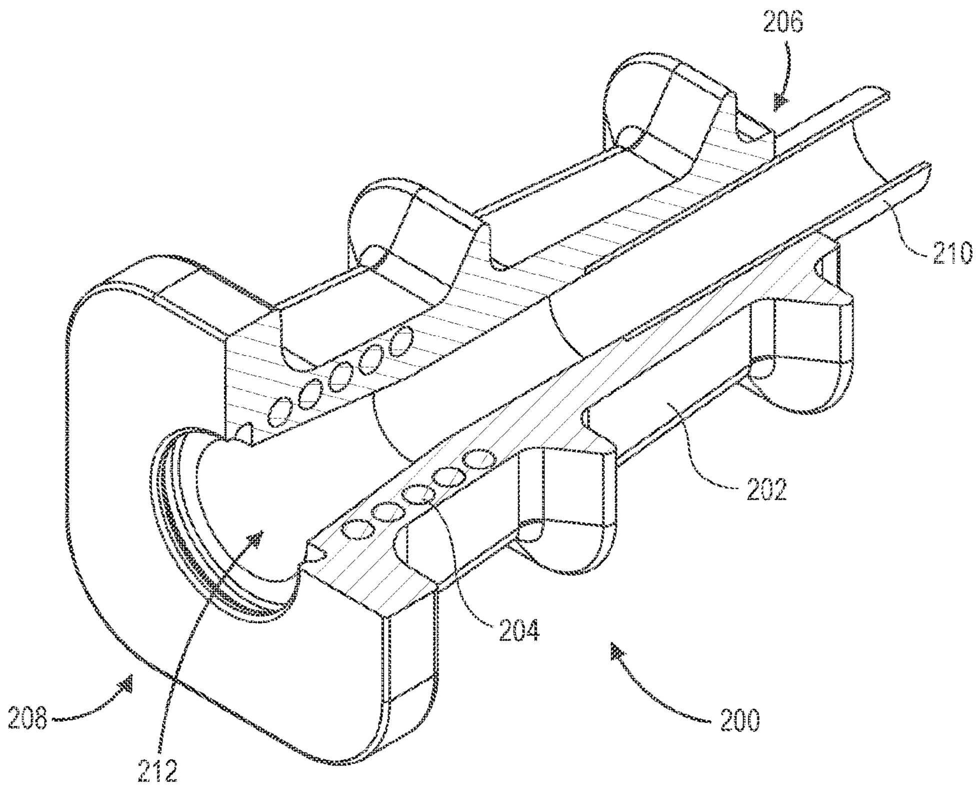

FIG. 5

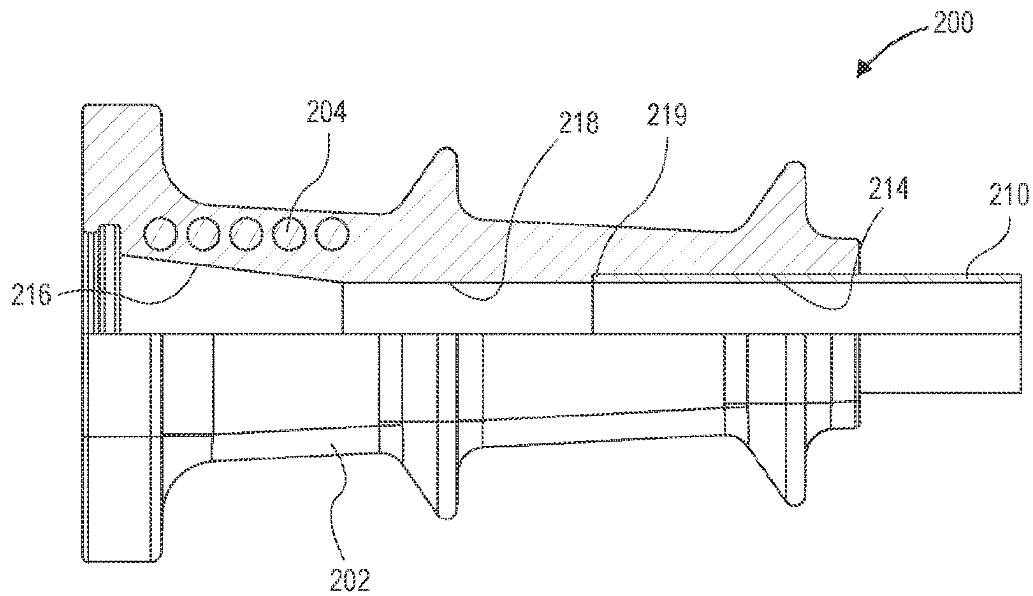

FIG. 6A

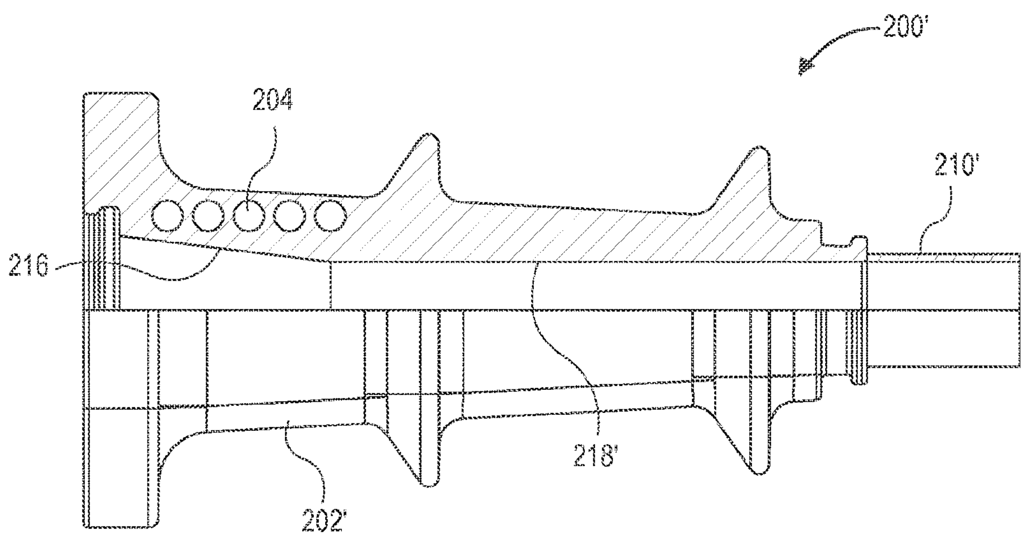

FIG. 6B

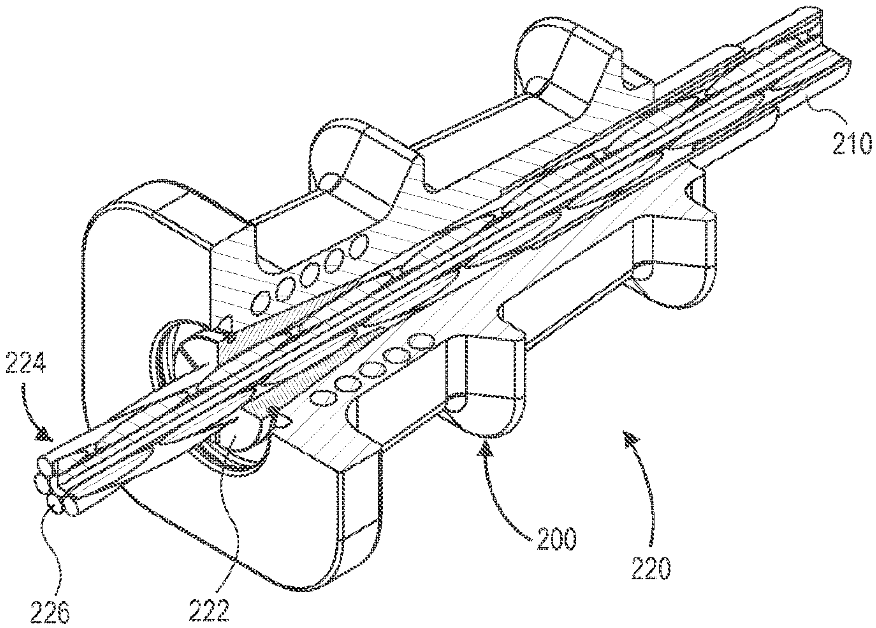

FIG. 7

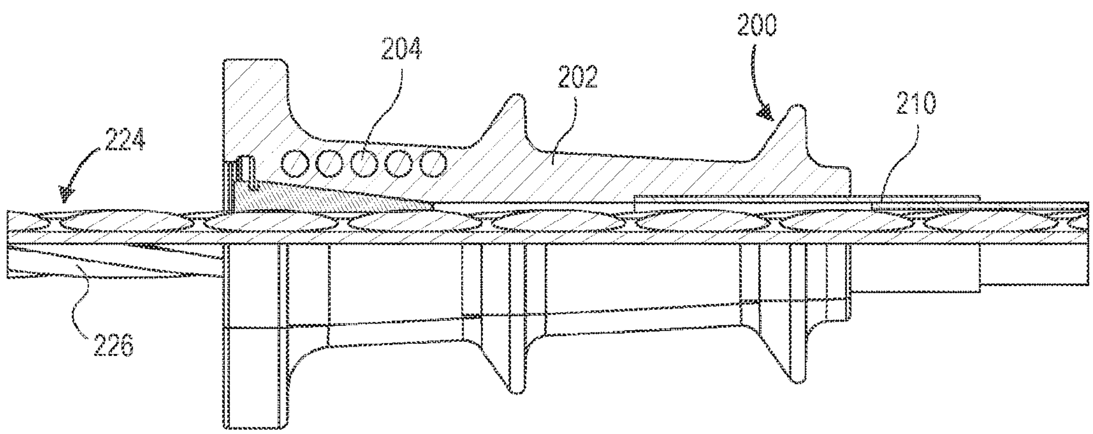

FIG. 8

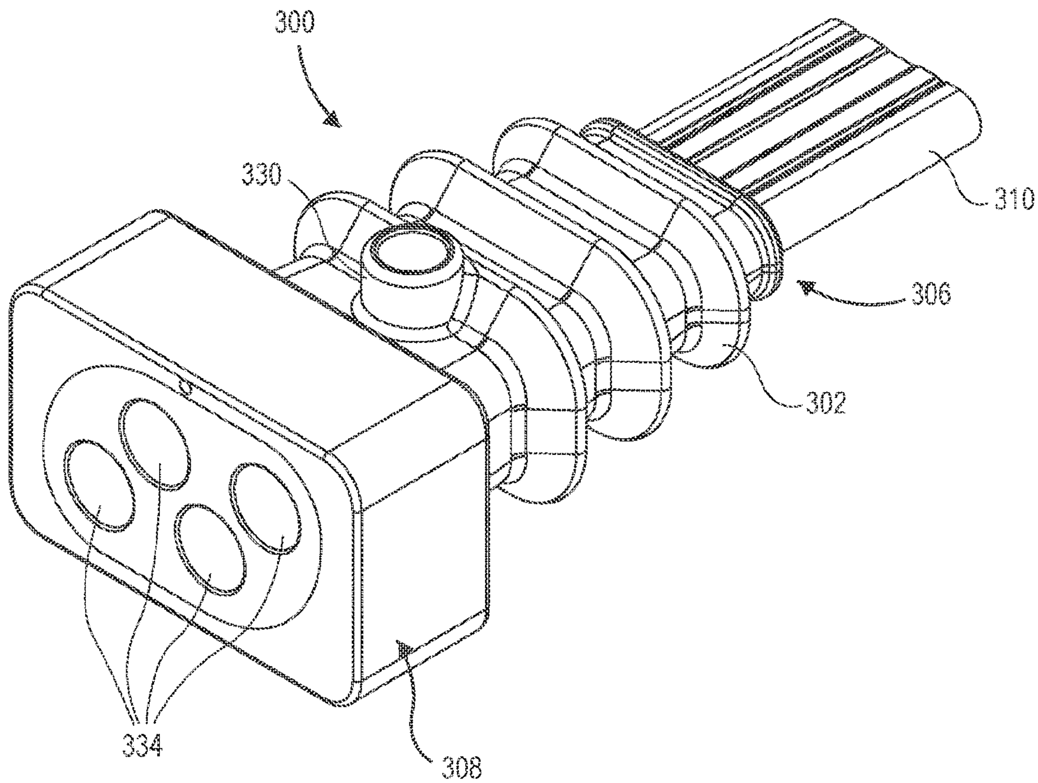

FIG. 9

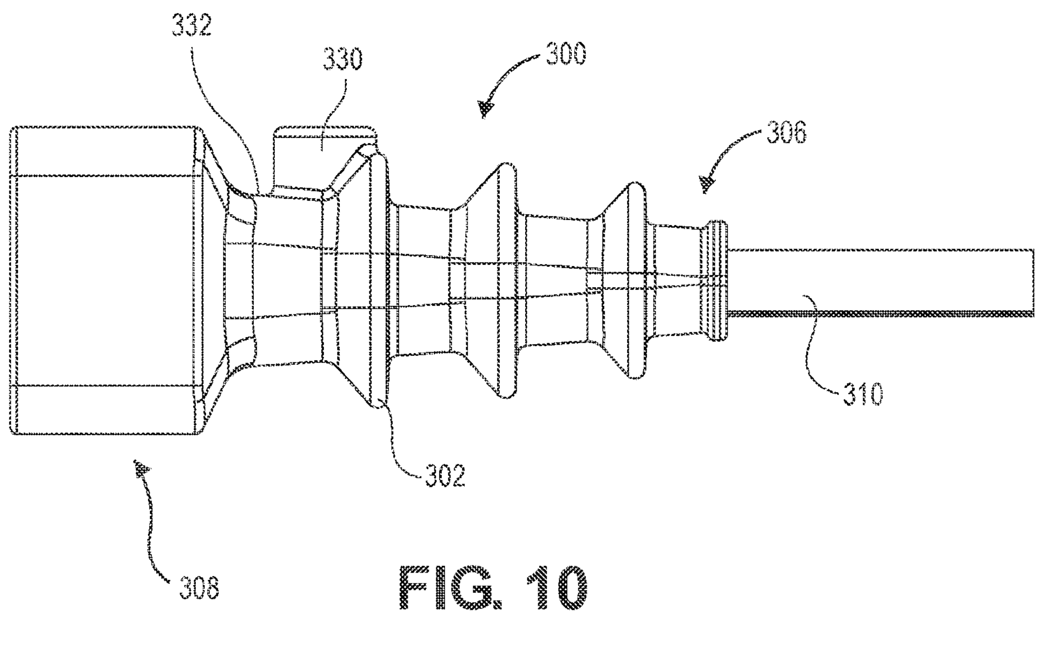

FIG. 10

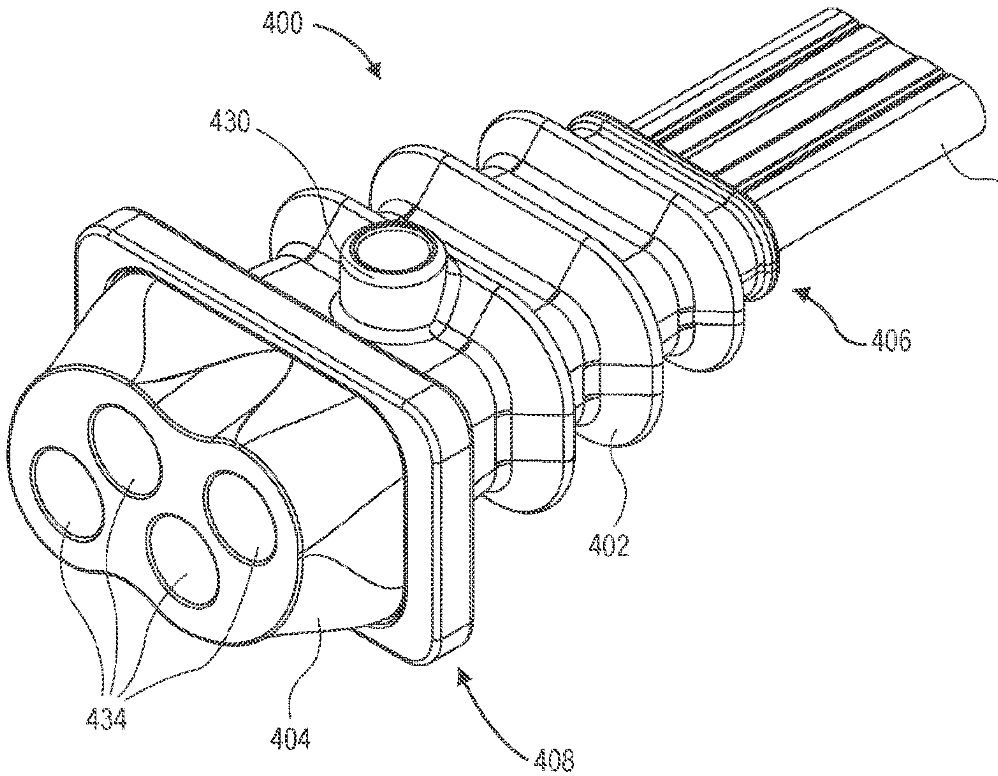

FIG. 11

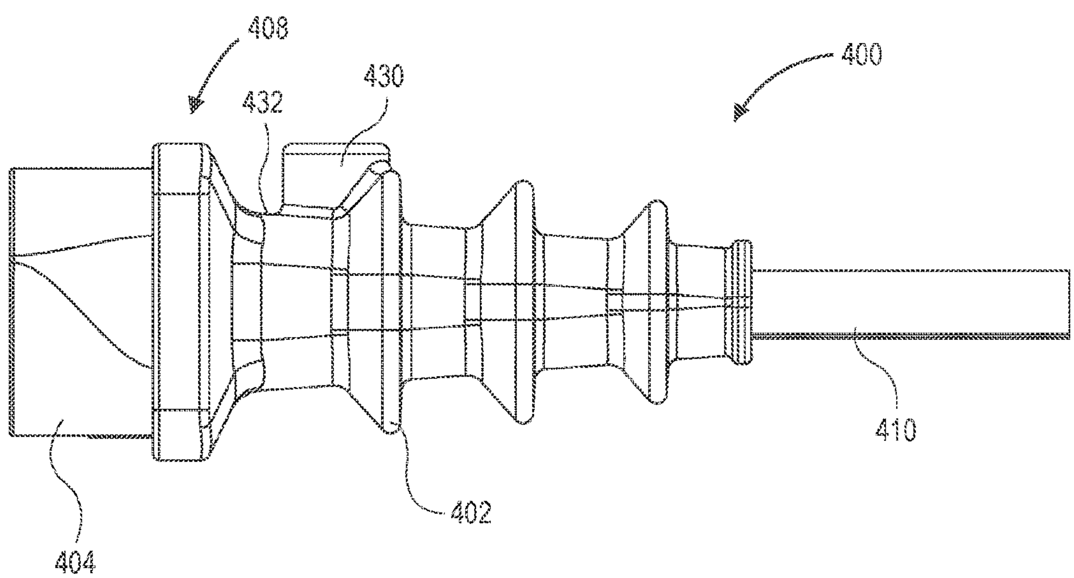

FIG. 12

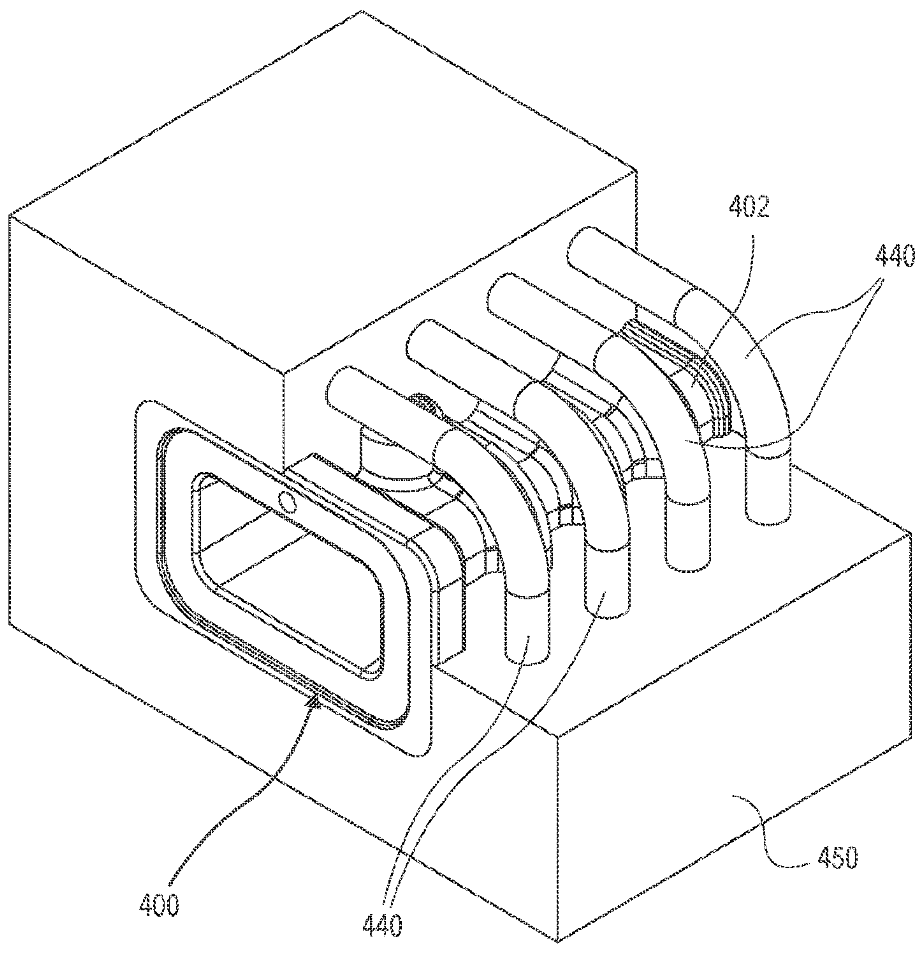

FIG. 13

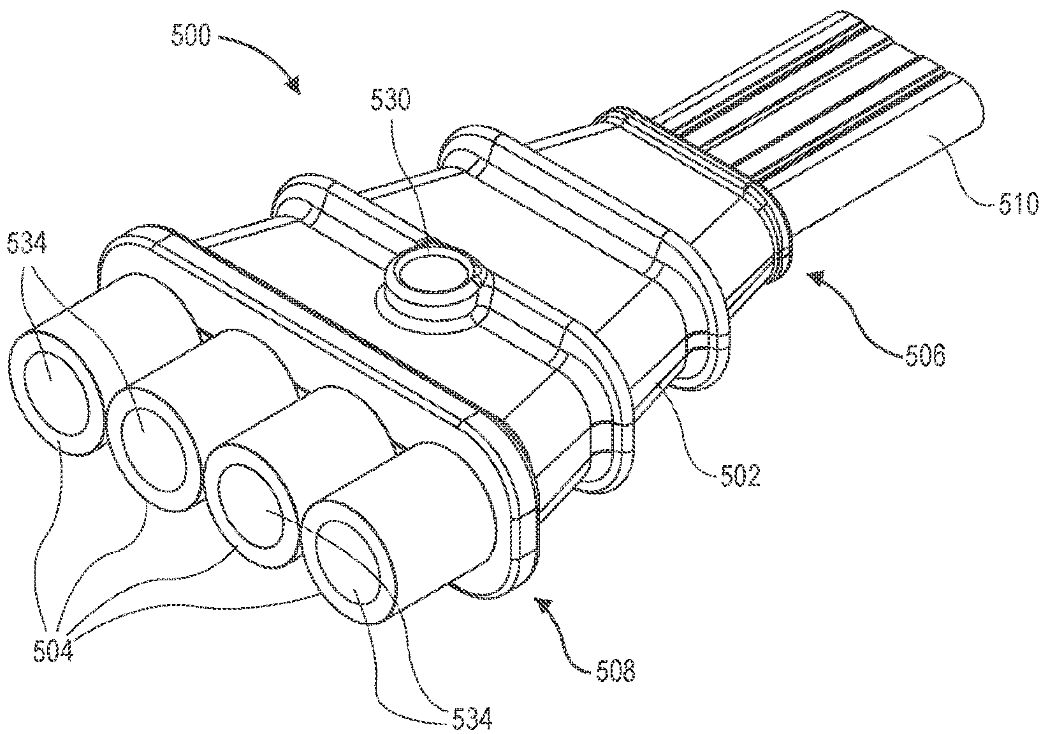
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FIG. 14

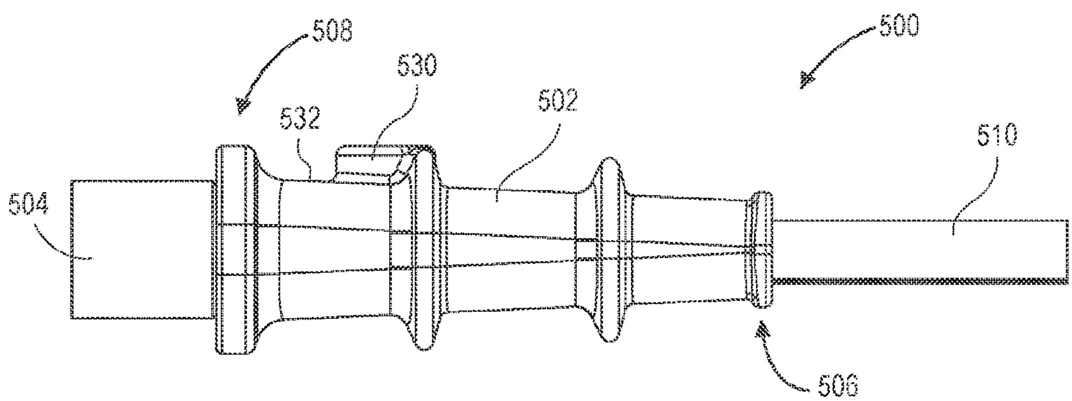
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FIG. 15

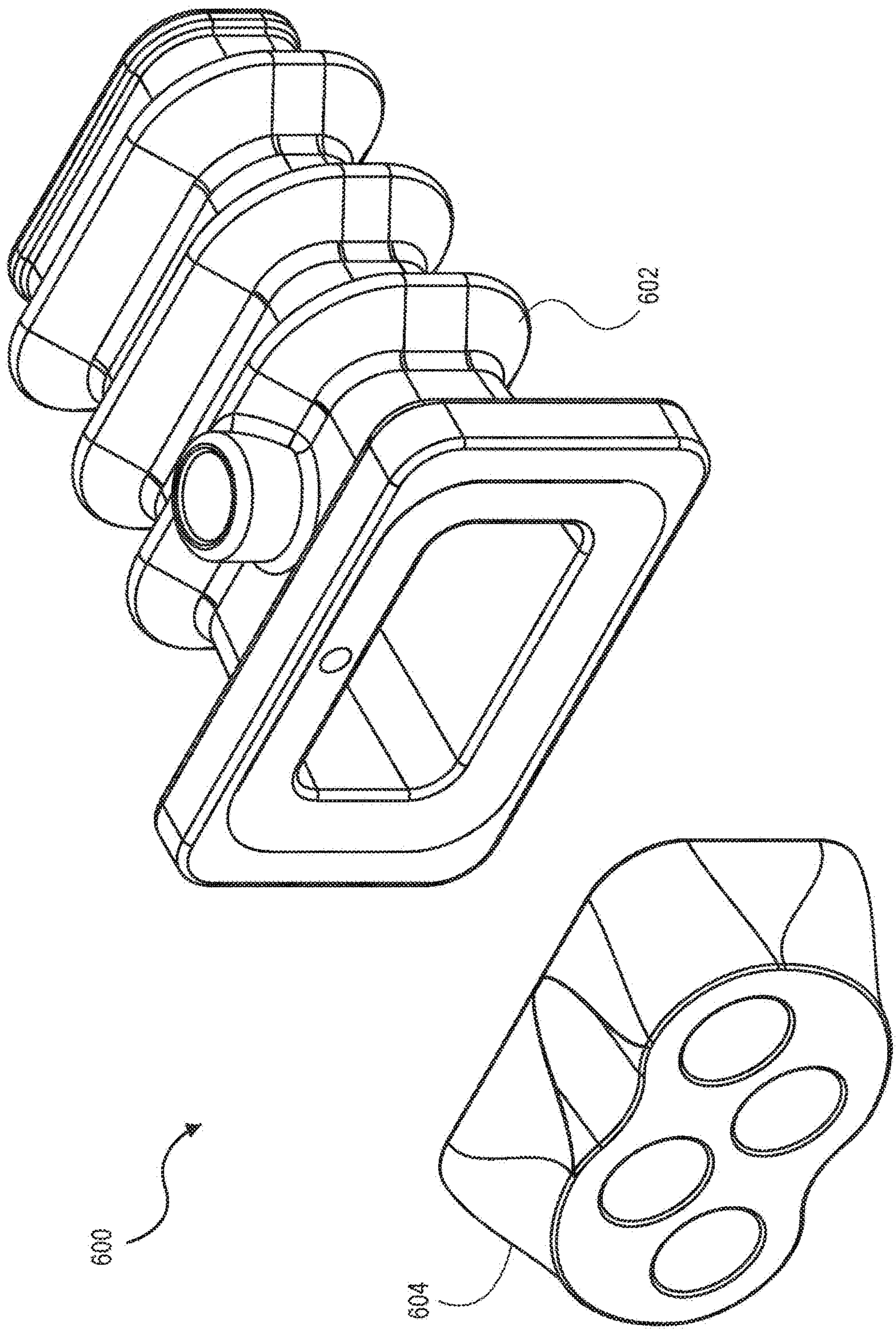

FIG. 16
PRIOR ART

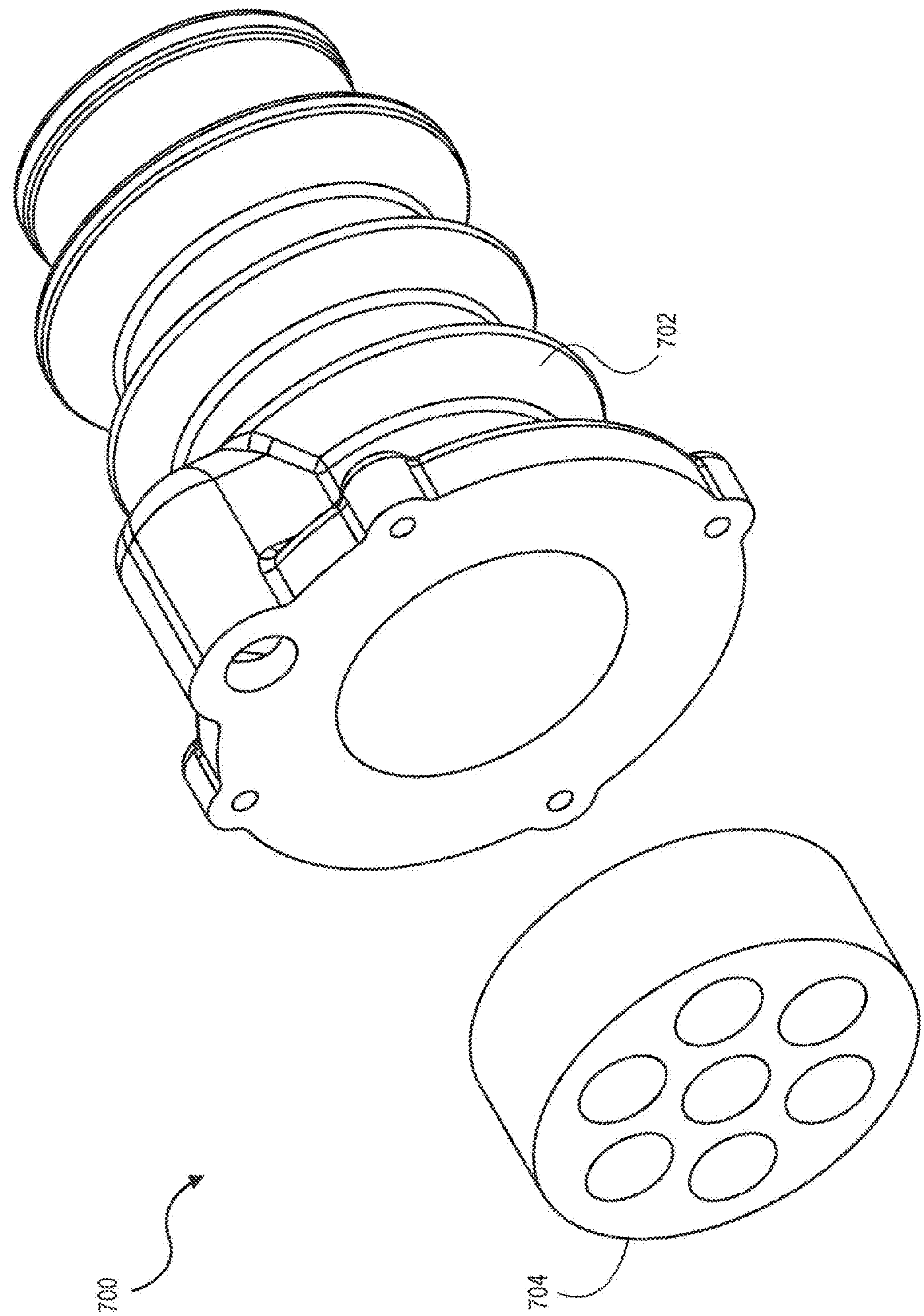

FIG. 17
PRIOR ART

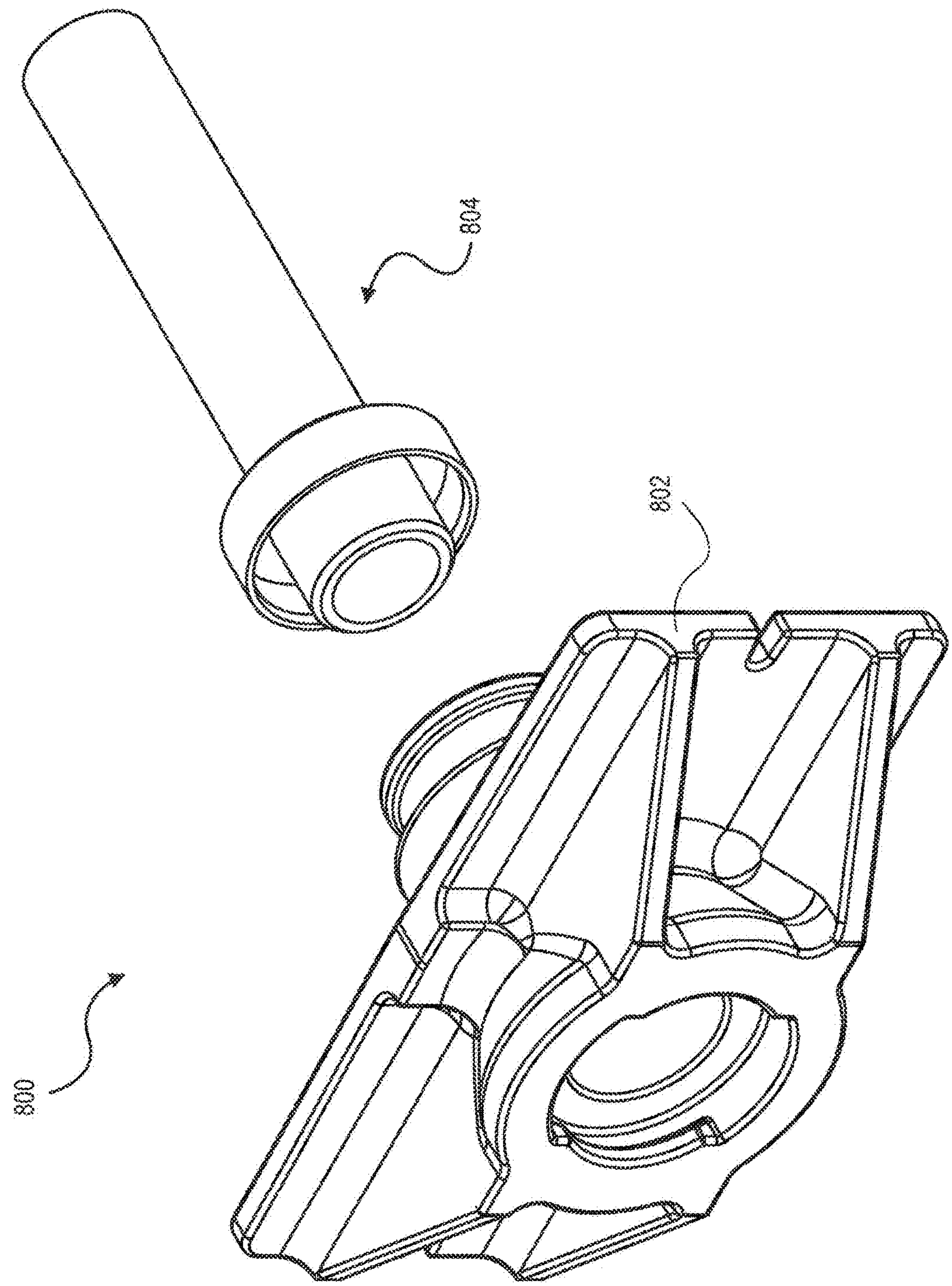

FIG. 18
PRIOR ART

CONCRETE POST-TENSIONING ANCHORS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a U.S. National Stage application of PCT/IB2021/054937 filed on Jun. 4, 2021, which claims priority to U.S. Provisional Application No. 63/052,283 filed on Jul. 15, 2020, the entire disclosures of which are hereby incorporated by reference in their entireties.

BACKGROUND

The present disclosure relates generally to post-tensioning anchors and, more particularly, to post-tensioning anchors made from concrete.

Prestressed concrete is a form of concrete used in construction, where the concrete is prestressed (compressed) during production such that the concrete is strengthened against tensile forces or stresses that will exist when the concrete is in use. The prestressing is produced by the tensioning of high-strength tension elements located within or adjacent to the concrete. Prestressed concrete has the characteristics of high-strength concrete when subject to any subsequent compression forces and of ductile high-strength steel when subject to tension forces. Thus, prestressed concrete has improved structural capacity and/or serviceability compared with conventionally reinforced concrete.

Post-tensioning is one type of prestressing where high-strength tension elements (e.g., steel cable) are placed before or after the concrete is cast. Then, after the concrete is cast and has gained strength, but typically before service loads are applied, the tension elements are pulled tight (i.e., tensioned) and anchored against the edges of the concrete (e.g, an outer edge or an edge in the middle of a slab). Post-tensioning may be carried out via monostrand systems, where each tension element is placed and stressed individually, or via multi-strand systems, where several tension elements are placed in a single conduit and where stressing can be done individually or simultaneously for the group.

In various types of construction applications (e.g., bridges, buildings, transfer beams, containment structures, other structural applications, other geotechnical foundations, and other civil applications), highly stressed tension elements are used in prestressed concrete construction or post-tensioned concrete construction or geotechnical engineering. The high tensile forces concentrated in the corresponding tension elements need to be dissipated by anchoring in the surrounding concrete substrate (e.g., prestressed concrete) structure or the ground.

Some conventional post-tensioning anchors are made from steel and/or iron casting. For example, some conventional multi-piece anchors for multistrand systems include a force transfer unit (or anchorage transfer guide) and at least one anchor head (or anchor block or wedge plate). The force transfer unit and the anchor head are made from steel and/or iron casting. Some conventional one-piece anchors, typically used for monostrand systems, include a force transfer unit that is made of steel or iron casting. In such conventional systems, the anchors need to be coated or encased with a corrosion resistant material to protect the anchors from corrosion.

FIGS. **16** and **17** illustrate conventional two-piece post-tensioning anchors **600**, **700** for multistrand systems. The anchors **600**, **700** include a force transfer unit **602**, **702** and an anchor head **604**, **704**. Tension elements (not shown) that extend through the force transfer unit **602**, **702** and the

anchor head **604**, **704** are locked in place by wedges (not shown) that are received by the openings in the anchor head **604**, **704**. The force transfer unit **602**, **702** and the anchor head **604**, **704** are made of steel or iron casting. FIG. **18** illustrates a conventional anchor **800** for a monostrand system. The anchor **800** includes a force transfer unit **802** and an extruder **804**. A tension element (not shown) that extends through the extruder **804** and the force transfer unit **802** is locked in place by a wedge (not shown) that is received by the opening in the force transfer unit **802**. The force transfer unit **802** is made of steel or iron casting.

It may be desirable to provide post-tensioning anchors and/or force transfer units for post-tensioning anchors that do not need to be coated or encased with a corrosion resistant material. More particularly, it may be desirable to provide post-tensioning anchors and/or force transfer units made from concrete. For example, it may be desirable to provide one-piece concrete anchors for monostrand and multistrand, bonded and unbonded systems. For example, it may be desirable to provide multi-piece anchors including a force transfer unit made from concrete and at least one anchor head made from steel or iron casting. For example, it may be desirable to provide one-piece or multi-piece concrete anchors with at least one steel member embedded in the anchor or close to the face of the anchor. It may also be desirable to provide post-tensioning anchor assemblies with reinforcement and without reinforcement in a concrete substrate at the anchor location.

SUMMARY

In accordance with various embodiments of the disclosure, a post-tensioning anchor is configured to post-tension at least one tension steel element that includes a plurality of steel wires. The post-tensioning anchor includes a force transfer unit configured to transmit prestressing force into a surrounding concrete substrate and at least one steel member configured to resist a force by the tension steel element on the force transfer unit.

In some aspects, the force transfer unit is made of high strength concrete, for example, high performance concrete or ultra-high-performance concrete.

According to various aspects, the force transfer unit is made of a concrete that contains organic, basalt, bare steel, stainless steel, or coated steel fibers.

According to some aspects, the force transfer unit has a first end and an opposite second end, and the first end being configured to receive at least one duct or connection piece or fitting and at least one tension steel element.

In some aspects, the at least one steel member is embedded in the force transfer unit at the second end. In various aspects, the steel member is a steel barrel, or a steel channel, while in other aspects, the steel member is a continuous steel spiral or a series of steel links. In some aspects, the steel members may include a combination of steel members placed anywhere along the force transfer unit, for example, embedded in or located outside the force transfer unit.

According to various aspects, the force transfer unit is configured to receive a plurality of tension steel elements, and the second end of the force transfer unit includes one bore or a number of bores corresponding to or greater than a number of the plurality of tension steel elements.

In various aspects, the at least one steel member is at least one anchor head disposed at the second end of the force transfer unit. In some aspects, the force transfer unit does not include a reinforcement member.

According to some aspects, the force transfer unit is configured to receive a plurality of tension steel elements, and the anchor head includes a number of bores corresponding to or greater than the number of the plurality of tension steel elements.

According to some aspects, the force transfer unit is configured to receive a plurality of tension steel elements, and a plurality of individual anchor heads corresponding to the number of the plurality of tension steel elements.

In some aspects, the concrete is high strength concrete having a compressive strength of at least 10,000 psi at 28 days, and preferably 15,000 to 50,000 psi at 28 days. According to various aspects, the high strength concrete has a tensile strength of greater than 400 psi.

In various aspects, the concrete can be any structural concrete including, but not limited to, conventional concrete, fiber reinforced concrete, self-compacting concrete, shrinkage-reducing concrete, or any combination thereof.

According to various aspects, an anchoring assembly includes one of the aforementioned post-tensioning anchors, at least one duct or connection fitting configured to receive the at least one tension steel element and being received by the post-tensioning anchor, and at least one clamping wedge configured to cooperate with the post-tensioning anchor to clamp the at least one tension steel element.

In various aspects, one of the aforementioned post-tensioning anchors is used in a construction application or a structural application including a concrete substrate with reinforcement surrounding the anchor in the concrete substrate.

According to some aspects, one of the aforementioned post-tensioning anchors is used in a construction application or a structural application including a concrete substrate without reinforcement surrounding the anchor in the concrete substrate.

In various aspects, the concrete anchors can take any shape such as circular, rectangular, oblong, multi-plane, or any combination thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments of the invention will now be further described, by way of example only, and with reference to the accompanying drawings, in which:

FIG. **1** is a perspective, partial cross-sectional view of an exemplary one-piece monostrand post-tensioning anchor in accordance with various aspects of the disclosure;

FIG. **2**A is a side, partial cross-sectional view of the post-tensioning anchor of FIG. **1**;

FIG. **2**B is a side, partial cross-sectional view of an exemplary one-piece monostrand post-tensioning anchor in accordance with various aspects of the disclosure:

FIG. **3** is a perspective, partial cross-sectional view of an exemplary anchoring assembly including the post-tensioning anchor of FIG. **1**;

FIG. **4** is a side, partial cross-sectional view of the exemplary anchoring assembly of FIG. **3**;

FIG. **5** is a perspective view of an exemplary one-piece monostrand post-tensioning anchor in accordance with various aspects of the disclosure;

FIG. **6**A is a side, partial cross-sectional view of the post-tensioning anchor of FIG. **5**;

FIG. **6**B is a side, partial cross-sectional view of an exemplary one-piece monostrand post-tensioning anchor in accordance with various aspects of the disclosure;

FIG. **7** is a perspective, partial cross-sectional view of an exemplary anchoring assembly including the post-tensioning anchor of FIG. **5**:

FIG. **8** is a side, partial cross-sectional view of the exemplary anchoring assembly of FIG. **7**;

FIG. **9** is a perspective view of an exemplary one-piece multistrand post-tensioning anchor in accordance with various aspects of the disclosure:

FIG. **10** is a side view of the post-tensioning anchor of FIG. **9**;

FIG. **11** is a perspective view of an exemplary two-piece multistrand post-tensioning anchor in accordance with various aspects of the disclosure;

FIG. **12** is a side view of the post-tensioning anchor of FIG. **11**;

FIG. **13** is a perspective view of the force transfer unit of the post-tensioning anchor of FIG. **11** with a schematic representation of reinforcement in a concrete substrate at the anchor location;

FIG. **14** is a perspective view of an exemplary multi-piece multistrand post-tensioning anchor in accordance with various aspects of the disclosure;

FIG. **15** is a side view of the post-tensioning anchor of FIG. **14**;

FIG. **16** is a perspective view of a conventional multi-piece multistrand post-tensioning anchor;

FIG. **17** is a perspective view of another conventional multi-piece multistrand post-tensioning anchor; and

FIG. **18** is a perspective view of a conventional one-piece monostrand post-tensioning anchor.

DETAILED DESCRIPTION OF EMBODIMENTS

FIGS. **1** and **2**A illustrate an exemplary one-piece monostrand post-tensioning anchor **100** in accordance with various aspects of the disclosure. The anchor **100** includes a force transfer unit **102** and at least one steel member **104**, for example, a steel barrel. It some aspects, the steel member **104** is embedded. The force transfer unit **102** has a first end **106** and an opposite second end **108**. The first end **106** is configured to receive a duct or connection fitting **110**, for example, an adaptor piece, a thin plastic sleeve, a sheet metal pipe, a steel duct, or a plastic duct. The connecting fitting **110** can be provided with the anchor **100** or installed at a later time. The barrel **104** is proximate the second end **108**.

The anchor **100** defines a through bore **112** having a first end bore portion **114**, a second end bore portion **116**, and a middle bore portion **118** between the first end bore portion **114** and the second end bore portion **116**. The first end bore portion **114** has an inner diameter that is greater than an inner diameter of the middle bore portion **118** such that a radially-extending wall **119** connects the first end bore portion **114** and the middle bore portion **118**. The radially-extending wall **119** faces toward the first end **106** of the force transfer unit **102**. The second end bore portion **116** has an inner diameter that tapers in a direction from the second end **108** of the force transfer unit toward the first end **106**.

It should be appreciated that in some embodiments, the first end bore portion **114** and the middle bore portion **118** conduit **110** may have the same diameter, thus eliminating the radially-extending wall **119**. In such an embodiment, the connection fitting **110** may extend all the way to the steel member **104**. In some embodiments, the conduit **110** may be omitted such that a duct abuts the first end **106** of the force transfer unit **102** (see, e.g., FIG. **6**B). In some embodiments, the connection fitting **110** may be eliminated, and tape (not shown) can instead be applied to a tension steel element that extends through the force transfer unit **102**.

Another exemplary one-piece monostrand post-tensioning anchor **100'** similar to anchor **100** is shown in FIG. **2**B. The anchor **100'** may include a main bore portion **118'** that extends from the end bore portion **116** to the first end **106** of the force transfer unit, and a duct or connection fitting **110'** may be configured to be disposed outside of the force transfer unit **102'** without entering the main bore portion **118'**.

The force transfer unit **102** is made of any structural concrete including, but not limited to, conventional concrete, fiber reinforced concrete, self-compacting concrete, shrinkage-reducing concrete, or any combination thereof. In some embodiments, the concrete is high strength concrete such as, but not limited to, high-performance concrete (HPC) or ultra-high-performance concrete (UHPC). High strength concrete has a compressive strength three to ten times or higher that of conventional concrete. Compressive strength is the ability of a material to resist a compression load. Conventional concrete used in structural applications typically has a compressive strength of 3,000 to 8,000 psi at 28 days, or 4,000 to 6,000 psi at 28 days. High strength concrete has a compressive strength of 10,000 to 50,000 psi or higher at 28 days. Another measure of strength is tensile strength or tension, which is how strong a material is when you pull it. While conventional concrete has a tensile strength of 100 to 400 psi, high strength concrete has a tensile strength of greater than 400 psi.

UHPC also includes durability properties of freeze/thaw resistance, chloride resistance (like in road salts), and abrasion resistance that are similar to hard rock. Freeze/thaw resistance is tested by subjecting concrete prisms to freezing and thawing while submerged in a water bath. UHPC exhibited low degradation reaching 100% of its material properties after 600 freeze/thaw cycles. In one aspect, chloride permeability is measured by ponding a 3-percent sodium chloride solution on the surface of the concrete for 90 days. After 90 days, the level of migration of chloride ions into the concrete is determined. UHPC showed extremely low chloride migration when tested, less than 10% the permeability of conventional concrete. In one aspect, abrasion resistance is determined by measuring the amount of concrete abraded off a surface by a rotating cutter in a given time period. UHPC demonstrates excellent abrasion resistance, nearly twice as resistant as conventional concrete. Thus, UHPC provides superior corrosion resistance in comparison with conventional concrete, therefore eliminating the need for coating to provide corrosion protection.

The ingredients of UHPC are mainly: cement, silica fume, fine quartz, sand, high-range water reducer, water, and fibers such as bare steel fibers, stainless steel fibers, coated steel fibers, polymer fibers, or organic fibers. In some aspects, steel fiber content is between 100 per cubic yard (pcy) and 500 pcy, or in some aspects 130 pcy to 350 pcy, or in some aspects larger than 400 pcy.

Referring now to FIGS. **3** and **4**, an exemplary anchoring assembly **120** including the post-tensioning anchor **100** is illustrated and described. The assembly **120** includes the anchor **100**, the connection fitting **110**, and one or more wedge pieces **122**, for example, clamping wedges. The connection fitting **110** is configured to encase a tension steel element **124**, for example, a steel strand which may include a plurality of twisted steel wires **126**. The tension steel element **124** may be coated with a protective coating. The prestressing tension steel element **124** can be manufactured as per the requirements of ASTM A-416 and/or ASTM A-779, or European Norm such as EN 10138, or equivalent international standards, and typical strand sizes include, but are not limited to, 0.50, 0.60, and 0.70 inch in diameter. A typical steel strand used for post-tensioning will have an ultimate strength of 270,000 psi, or between 270,000 psi and 320,000 psi.

As shown in FIGS. **3** and **4**, the tension steel element **124** is fed through the through bore **112** in the anchor **100** from either end of the anchor **100**. The first end **106** of the force transfer unit **102** is configured to receive the connection fitting or a conduit **110**, and the radially-extending wall **119** is configured to limit a distance that the connection fitting **110** can be inserted into the through bore **112**. Alternatively, the first end **106** of the force transfer unit **102** may include internal threads or any other conventional means for limiting the distance that the connection fitting **110** can be inserted into the through bore, and extending bore portion **118** to the first end **106**. The second end **108** of the force transfer unit **102** is configured to receive the one or more wedge pieces **122**, and the one or more wedge pieces **122** are configured to cooperate with the tapered second end bore portion **116** such that a tensile force of the tension steel element **124** is introduced into the one or more wedge pieces **122**, which are pressed axially into the tapered second end bore portion **116** of the anchor **100** and introduce the tensile force via their outer surface into the anchor **100**. The steel member **104** is configured to resist the force at the one or more wedge pieces **122**. It should be appreciated that in some embodiments, the conduit **110** may extend all the way to the steel member **104**, and in some embodiments, the conduit **110** may be omitted such that a duct will abut the first end **106** of the force transfer unit **102** (see, e.g., FIG. **6**B).

FIGS. **5** and **6**A illustrate another exemplary one-piece monostrand post-tensioning anchor **200** in accordance with various aspects of the disclosure. The anchor **200** is similar to the anchor **100** described above, but includes some variations as described below. The anchor **200** includes a force transfer unit **202** and at least one steel member **204**, for example, a steel spiral or a series of steel links. The force transfer unit **202** has a first end **206** and an opposite second end **208**. The first end **206** is configured to receive a duct or a connection fitting **210**, for example, an adaptor piece, a thin plastic sleeve, a sheet metal pipe, a steel duct, or a plastic duct. The steel member **204** is proximate the second end **208**. The force transfer unit **202** is made of any structural concrete including, but not limited to, conventional concrete, fiber reinforced concrete, self-compacting concrete, shrinkage-reducing concrete, or any combination thereof. In some embodiments, the concrete is high strength concrete such as, but not limited to, HPC or UHPC.

The anchor **200** defines a through bore **212** having a first end bore portion **214**, a second end bore portion **216**, and a middle bore portion **218** between the first end bore portion **214** and the second end bore portion **216**. The first end bore portion **214** has an inner diameter that is greater than an inner diameter of the middle bore portion **218** such that a radially-extending wall **219** connects the first end bore portion **214** and the middle bore portion **218**. The radially-extending wall **219** faces toward the first end **206** of the force transfer unit **202**. Alternatively, the first end **206** of the force transfer unit **202** may include internal threads or any other conventional means for limiting the distance that the connection fitting **210** can be inserted into the through bore, and extending bore portion **218** till the first end **206**. The second end bore portion **216** has an inner diameter that tapers in a direction from the second end **208** of the force transfer unit toward the first end **206**. It should be appreciated that in some embodiments, the connection fitting **210** may extend all the way to the end of the middle bore portion **218** adjacent the second end bore portion **216**.

Another exemplary one-piece monostrand post-tensioning anchor **200'** similar to anchor **200** is shown in FIG. **6**B. The anchor **200'** may include a main bore portion **218'** that extends from the end bore portion **216** to the first end **206** of the force transfer unit, and a duct or connection fitting **210'** may be configured to be disposed outside of the force transfer unit **202'** without entering the main bore portion **218'**.

Referring now to FIGS. **7** and **8**, an exemplary anchoring assembly **220** including the post-tensioning anchor **200** is illustrated and described. The assembly **220** includes the anchor **200**, the connection fitting **210**, and one or more wedge pieces **222**, for example, clamping wedges. The connection fitting **210** is configured to encase a steel tension element **224**, for example, a steel strand, which may include a plurality of twisted steel wires **226**. The steel tension element **224** may be coated with a protective coating. The prestressing steel tension element **224** can be manufactured as per the requirements of ASTM A-416 and ASTM A-779, or equivalent international standards, and typical strand sizes include, but are not limited to, 0.50, 0.60, and 0.70 inch in diameter. A typical steel strand used for post-tensioning will have an ultimate strength of 270,000 psi or between 270,000 psi and 320,000 psi.

As shown in FIGS. **7** and **8**, the tension steel element **224** is fed through the through bore **212** in the anchor **200** from either end of the anchor **200**. The first end **206** of the force transfer unit **202** is configured to receive the connection fitting **210**, and the radially-extending wall **219** is configured to limit a distance that the connection fitting **210** can be inserted into the through bore **212**. Alternatively, the first end **206** of the force transfer unit **202** may include internal threads or any other conventional means for limiting the distance that the connection fitting **210** can be inserted into the through bore and extending bore portion **218** to the first end **206**. The second end **208** of the force transfer unit **202** is configured to receive the one or more wedge pieces **222**, and the one or more wedge pieces **222** are configured to cooperate with the tapered second end bore portion **216** such that a tensile force of the tension steel element **224** is introduced into the one or more wedge pieces **222**, which are pressed axially into the tapered second end bore portion **216** of the anchor **200** and introduce the tensile force via their outer surface into the anchor **200**. The steel member **204** is configured to resist the force at the one or more wedge pieces **222**.

FIGS. **9** and **10** illustrate an exemplary one-piece multistrand post-tensioning anchor **300** in accordance with various aspects of the disclosure. The anchor **300** is similar to the anchors **100**, **200** described above, but includes some variations as described below. The anchor **300** includes a force transfer unit **302** and a steel member **335** configured to resist the clamping force. The force transfer unit **302** is made of any structural concrete including, but not limited to, conventional concrete, fiber reinforced concrete, self-compacting concrete, shrinkage-reducing concrete, or any combination thereof. In some embodiments, the concrete is high strength concrete such as, but not limited to, HPC or UHPC.

The force transfer unit **302** has a first end **306** and an opposite second end **308**. The first end **306** is configured to receive at least one connection fitting **310** or a duct, for example, an adaptor piece, a thin plastic sleeve, a sheet metal pipe, a steel duct, or a plastic duct. The steel member is in the force transfer unit **302** proximate the second end **308**. The force transfer unit **302** may include a port **330** in an outer wall **332**. The port **330** is configured to receive a grout tube (not shown) that is configured to direct grout into the through bore and duct **310**.

The anchor **300** defines a through bore (not shown) configured to receive a plurality of tension steel elements from either end of the anchor **300**. The second end **308** of the force transfer unit **302** includes a plurality of bores **334**, each configured to receive one of the plurality of tension steel elements fed through the through bore in the anchor **300**. Each of the plurality of bores **334** at the second end **308** of the force transfer unit **302** is configured to receive one or more wedge pieces similar to the wedge pieces **122**, **222** described above. The one or more wedge pieces are configured to cooperate with a tapered inner wall of a respective bore **334** at the second end **308** of the force transfer unit **302** such that a tensile force of the tension steel elements is introduced into the one or more wedge pieces, which are pressed axially into the tapered inner wall of a respective bore **334** and introduce the tensile force via their outer surface into the anchor **300**.

FIGS. **11** and **12** illustrate an exemplary two-piece multistrand post-tensioning anchor **400** in accordance with various aspects of the disclosure. The anchor **400** is similar to the anchors **100**, **200**, **300** described above, but includes some variations as described below. The anchor **400** includes a force transfer unit **402** and an anchor head **404** configured to resist the clamping force. The force transfer unit **402** is made of any structural concrete including, but not limited to, conventional concrete, fiber reinforced concrete, self-compacting concrete, shrinkage-reducing concrete, or any combination thereof. In some embodiments, the concrete is high strength concrete such as, but not limited to, HPC or UHPC. The anchor head **404** is made of steel or iron casting.

The force transfer unit **402** has a first end **406** and an opposite second end **408**. The first end **406** is configured to receive at least one connection fitting **410**, for example, an adaptor piece, a thin plastic sleeve, a sheet metal pipe, a steel duct, a plastic duct, or any conduit. The anchor head **404** is disposed at the second end **408** of the force transfer unit **402**.

The anchor **400** defines a through bore (not shown) or a plurality of through bores (not shown) configured to receive a plurality of tension steel elements. The force transfer unit **402** may include a port **430** in an outer wall **432**. The port **430** is configured to receive a grout tube (not shown) that is configured to direct grout into the through bore and the connection fitting **410**.

The anchor head **404** includes a plurality of bores **434**, each configured to receive one of the plurality of tension steel elements fed through the through bore in the force transfer unit **402** from either end of the anchor **400**. The bores **434** in the anchor head **404** are configured to receive one or more wedge pieces similar to the wedge pieces **122**, **222** described above. The one or more wedge pieces are configured to cooperate with a tapered inner wall of a respective bore **434** in the anchor head **404** such that a tensile force of the tension steel elements is introduced into the one or more wedge pieces, which are pressed axially into the tapered inner wall of a respective bore **434** and introduce the tensile force via their outer surface into the anchor head **404**.

The high strength concrete force transfer unit **402** may not require any steel member and may not need to be coated or encapsulated to provide protection against corrosion. In some construction applications, the anchor **400** can be deployed without reinforcement in the surrounding concrete substrate. In other applications, the anchor **400** can be deployed with reinforcement **440** (i.e., steel reinforcement) in the surrounding concrete substrate **450**, as shown in FIG. **13**.

FIGS. **14** and **15** illustrate an exemplary multi-piece multistrand post-tensioning anchor **500** in accordance with various aspects of the disclosure. The anchor **500** is similar to the anchors **100**, **200**, **300**, **400** described above, but includes some variations as described below. The anchor **500** includes a force transfer unit **502** and a plurality of anchor heads or barrels **504** configured to resist the clamping force. The force transfer unit **502** is made of any structural concrete including, but not limited to, conventional concrete, fiber reinforced concrete, self-compacting concrete, shrinkage-reducing concrete, or any combination thereof. In some embodiments, the concrete is high strength concrete such as, but not limited to, HPC or UHPC. The anchor heads or barrels **504** are made of steel or iron casting.

The force transfer unit **502** has a first end **506** and an opposite second end **508**. The first end **506** is configured to receive at least one connection fitting **510**, for example, an adaptor piece, a thin plastic sleeve, a sheet metal pipe, a steel duct, or a plastic duct. The anchor barrels **504** are disposed at the second end **508** of the force transfer unit **502**.

The anchor **500** defines a through bore (not shown) or a plurality of through bores (not shown) configured to receive a plurality of tension steel elements. The force transfer unit **502** may include a port **530** in an outer wall **532**. The port **530** is configured to receive a grout tube (not shown) that is configured to direct grout into the through bore and duct **510**.

Each of the anchor barrels **504** includes a respective bore **534** configured to receive one of the plurality of tension steel elements fed through the through bore in the force transfer unit **502** from either end of the anchor **500**. The bores **534** in the anchor barrels **504** are configured to receive one or more wedge pieces similar to the wedge pieces **122**, **222** described above. The one or more wedge pieces are configured to cooperate with a tapered inner wall of a respective bore **534** in the anchor barrels **504** such that a tensile force of the tension steel elements is introduced into the one or more wedge pieces, which are pressed axially into the tapered inner wall of a respective bore **534** and introduce the tensile force via their outer surface into the anchor head **504**.

The high strength concrete force transfer unit **502** may not require any steel member and may not need to be coated or encapsulated to provide protection against corrosion. In some construction applications, and/or structural applications, and/or geotechnical applications, the anchor **500** can be deployed in bonded or unbonded post-tensioned concrete without reinforcement in the surrounding concrete substrate. In other applications, the anchor **500** can be deployed in bonded or unbonded post-tensioned concrete with reinforcement in the surrounding concrete substrate, similar to that shown in FIG. **13**. Likewise, anchors **100**, **200**, **300** can be deployed without reinforcement in the surrounding concrete or with reinforcement in the surrounding concrete substrate.

Additional embodiments include any one of the embodiments described above, where one or more of its components, functionalities, or structures is interchanged with, replaced by or augmented by one or more of the components, functionalities, or structures of a different embodiment described above.

Although several embodiments of the disclosure have been disclosed in the foregoing specification, it is understood by those skilled in the art that many modifications and other embodiments of the disclosure will come to mind to which the disclosure pertains, having the benefit of the teaching presented in the foregoing description and associated drawings. It is thus understood that the disclosure is not limited to the specific embodiments disclosed herein above, and that many modifications and other embodiments are intended to be included within the scope of the appended claims. Moreover, although specific terms are employed herein, as well as in the claims which follow, they are used only in a generic and descriptive sense, and not for the purposes of limiting the present disclosure, nor the claims which follow.

What is claimed is:

1. An anchoring assembly configured to post-tension at least one tension steel element, the anchoring assembly comprising:

a post-tensioning anchor comprising:

a force transfer unit comprising a first end, a second end, and a bore extending therethrough, wherein the bore comprises a second end bore portion proximate to the second end, wherein the second end bore portion has a diameter that increases proceeding toward the second end, wherein the force transfer unit is configured to transmit a prestressing force into a surrounding concrete substrate, and wherein the force transfer unit is made of concrete; and

at least one steel member positioned proximate to the force transfer unit;

a connection fitting positioned at least partially in the bore and proximate to the first end, wherein the connection fitting is between the at least one tension steel element and the force transfer unit; and

at least one wedge piece positioned at least partially in the bore and proximate to the second end, wherein the at least one steel member is outward from the at least one wedge piece, wherein the at least one wedge piece cooperates with the second end bore portion such that a post-tensioning force of the at least one tension steel element is introduced into the at least one wedge piece, which is pressed axially into the second end bore portion and introduces the post-tensioning force into the post-tensioning anchor, and wherein the at least one steel member is configured to partially resist the post tensioning force at the at least one wedge piece.

2. The anchoring assembly of claim **1**, wherein the first end is configured to receive the at least one tension steel element,

wherein the force transfer unit is configured to permit the at least one tension steel element to extend through the force transfer unit, and

wherein the second end is configured to receive the at least one wedge to clamp the force transfer unit with the at least one tension steel element.

3. The anchoring assembly of claim **2**, wherein the at least one steel member is at the second end of the force transfer unit.

4. The anchoring assembly of claim **3**, wherein the at least one steel member is embedded in the force transfer unit.

5. The anchoring assembly of claim **3**, wherein the at least one steel member is a steel conduit.

6. The anchoring assembly of claim **5**, wherein the steel conduit is a steel barrel or a steel channel.

7. The anchoring assembly of claim **3**, wherein the at least one steel member is a steel spiral or a series of steel links.

8. The anchoring assembly of claim **2**, wherein the at least one steel member is a plurality of steel conduits and/or steel spirals and/or steel links placed anywhere between the first end and the second end.

9. The anchoring assembly of claim **2**, wherein the plurality of steel conduits and/or steel spirals and/or steel links are located at at least two different locations between the first end and the second end.

10. The anchoring assembly of claim **2**, wherein the force transfer unit is configured to receive a plurality of the at least one tension steel element, and the second end of the force transfer unit includes one bore or a number of bores corresponding to or greater than a number of the plurality of the at least one tension steel element.

11. The anchoring assembly of claim **1**, wherein the force transfer unit is made of a concrete that contains but not limited to organic, basalt, bare steel, or coated steel fibers.

12. The anchoring assembly of claim **1**, wherein the at least one steel member is a plurality of steel conduits and/or steel spirals and/or steel links.

13. The anchoring assembly of claim **1**, wherein the at least one steel member is coated with a material configured to protect the steel member from corrosion.

14. The anchoring assembly of claim **1**, wherein the force transfer unit is configured to receive a plurality of the at least one tension steel element, and the anchor head includes a number of bores corresponding to or greater than a number of the plurality of the at least one tension steel element.

15. The anchoring assembly of claim **1**, wherein the concrete has a minimum compressive strength of at least 10,000 psi.

16. The anchoring assembly of claim **1**, wherein the concrete has a minimum compressive strength of 15,000 to 50,000 psi.

17. The anchoring assembly of claim **1**, wherein the concrete has a tensile strength larger than 400 psi.

18. The anchoring assembly of claim **1**, wherein the wedge piece presses radially into the at least one steel member.

19. The anchoring assembly of claim **1**, wherein the wedge piece does not press radially into the force transfer unit.

20. The anchoring assembly of claim **1**, wherein the at least one steel member is in the force transfer unit.

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